

Validation (Deterministic Python)

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Purpose

The Validation module runs eight independent checks on every generated plan. The pipeline is **strictly deterministic and AI-free**: it uses scikit-learn TF-IDF and rule-based comparisons only. This separation from the generation step is a hard requirement from the SRS (Section 1.2) - the validator cannot be the same intelligence that produced the artefact.

Why this matters

Two independent pipelines - one to generate, one to check - is how SkillsPrint claims trustworthiness. If the generator invents a rule, hallucinates a page number, or leaves a mandatory requirement uncovered, the validator catches it before the plan reaches the employee.

Check	What it measures	Failure surface
Coverage	Percentage of the role's mandatory requirements covered by the plan	Uncovered items listed by code
Traceability	Every task/quiz/assessment carries a valid source document citation and flagged	Items without citation and flagged
Hallucination	Cosine similarity between plan text and the cited sources	Low similarity items flagged
Duplicates	Similar tasks or quizzes appearing multiple times	Duplicate pairs listed
Contradiction	Plan asserting X while a cited chunk asserts not X	Contradiction pairs listed
Role relevance	Plan content matches the target job role's description	Off-role and off-plan flagged
Sequence	Prerequisites appear before their dependents	Out-of-order requirements listed
Distractors	Quiz distractors are non-obvious and non-repetitive	Weak distractors listed

SRS verification statuses

- **verified** - all eight checks pass at the strict threshold.
- **verified_with_warning** - all pass, one or more with borderline scores.
- **partially_verified** - one check fails but the failure is below the escalation threshold.
- **coverage_gap** - one or more mandatory requirements are not covered.
- **hallucination_detected** - one or more items fail traceability similarity.
- **contradiction_detected** - two cited sources are contradictory.
- **duplicate_detected** - similar tasks or quizzes repeat.
- **role_mismatch** - the plan drifts outside the role's scope.
- **sequence_issue** - prerequisites are ordered incorrectly.
- **failed** - two or more checks fail hard; the plan is not releasable.